The use of similarity measures to hydraulic systems condition monitoring

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Abstract

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1. Introduction

Hydraulic systems (i.e., a type of mechanical system that uses pressurized fluid to transmit and amplify forces) are widely employed across diverse industrial production processes. Nevertheless, complexity of their system structure presents challenges in developing a maintenance strategy [1]. In fact, hydraulic systems are used in a variety of applications, including construction equipment, aircraft, automobiles, and industrial machinery. The downtime of these systems results in significant financial costs. Therefore, preventive maintenance aimed at minimizing costs proves to be essential.

In this problematic scenario, several authors propose predictive models involving classical methods and methods with machine learning to estimate the remaining useful life of equipment or the status of each system component based on similarity: Dynamic Time Warping, Autoencoder, Siamese and Triplet networks [2, 3, 4].

The dynamic time warping was utilized for measuring the similarity between datas with time distortion. In [2],

In this context, this paper is dedicated to applying the main similarity measures in a hydraulic system to obtain information about the system's stability and cooling condition. Furthermore, it aims to optimize the performance of these measures, determining the most suitable measure to be applied in an industrial scenario, i.e., taking into account the complexity of the models.

2. Data Split

In this section, it discuss the data split process used for training and testing the models.

2.1. Hold-out

Hold-out is a simple and common way for validate the performance of machine learning models. In this data split, the dataset is splitting up into a training, validation and testing set randomly, being used to see how the model performs on unseen data, also it's possible to observe the phenomenon of over fitting by a large number of epochs by the presence of validation set.

2.2. K-fold cross-validation

The formal way of split up the set is further split into the training set and the validation set for the model selection [5]. Similarly, the dataset is split up into three sets randomly: training, validation and testing sets. However, only the testing set is fixed, i.e., the dataset is split up randomly into two sets, one of them is directed to the testing set and the other is split up into k sets randomly, then each one of the k sets becomes the validation and the remaining k-1 becomes the training set, as shown in the Figure 1. Naturally, the evaluation parameter is required to be the average of the k interactions of each evaluation measure with different setting.

3. Data preprocessing

Input normalization is the basic step in data preparation, which can facilitate the subsequent data processing and accelerate the convergence of the models, as cited by [5]. So, in this paper, the effect of four normalization methods on the performance of proposed models are discussed.

Given inputs of the form $x = (x_1, \cdots, x_n) \in \mathbb{R}^n$:

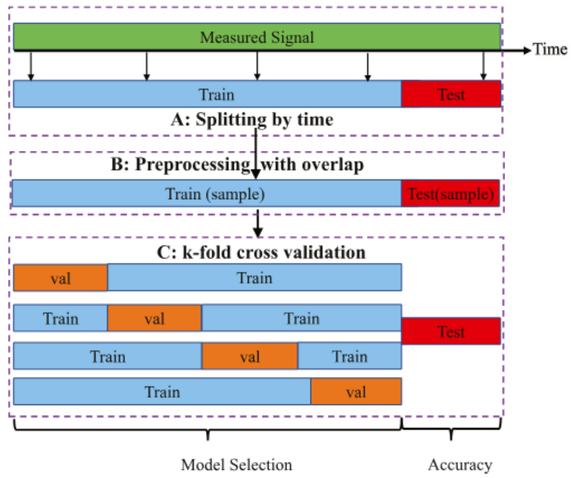

Figure 1: Random data splitting strategy.

3.1. Global Normalization Methods

Define normalization methods such as

$$x_i^{normalize-1} = \gamma_1 + \gamma_2 * \frac{x_i - x_i^{glob-min}}{x_i^{glob-max} - x_i^{glob-min}}, \quad i = 1, \cdots, n,$$

where γ_j is a constant to be defined for $j \in \{1, 2\}$, $x_i^{glob-min}$ and $x_i^{glob-max}$ are the global minimum and maximum of all selected sample values in x_i, respectively.

If $(\gamma_1, \gamma_2) = (0, 1)$, then it denotes as Global Maximum-Minimum Normalization. On the other hand, if $(\gamma_1, \gamma_2) = (-1, 2)$, then it denotes as Global [-1, 1] Normalization.

3.2. Local Normalization Methods

3.2.1. Local Decimal Scaling Normalization

This normalization method can be implemented by

$$x^{normalize-3} = \frac{x}{10^j},$$

where j is the $\log_{10}$ maximum absolute of the set $\{x_i\}_{i=1}^n$.

3.2.2. Local z-score Normalization

This normalization method can be implemented by

$$x^{normalize-4} = \frac{x - X^{mean}}{X^{std}},$$

where X^{mean} and X^{std} are the mean value and the standard deviation of the set $\{x_i\}_{i=1}^{n}$, respectively.

4. Similarity measures

4.1. Dynamic Time Warping - DTW

Dynamic time warping is a dynamic programming approach to align time-series with shifting and distortion in time, also its popularity by be extremely efficient as similarity measure between time-series, as seen in [6, 7, 2].

Given a distance function d' and two time series, $X = (x_{t_0}, x_{t_1}, \cdots, x_{t_{N-1}})$, $N \in \mathbb{N}$ and $Y = (y_{t_0}, y_{t_1}, \cdots, y_{t_{M-1}})$, $M \in \mathbb{N}$, build a local distance matrix $C \in \mathbb{R}^{N\times M}$ defined by:

$$c_{i,j} = d'(x_{t_i}, y_{t_j}), i \in \{0, \cdots, N-1\},\ j \in \{0, \cdots, M-1\}\ .$$

Then, build the global cost matrix $D \in \mathbb{R}^{(N+1)\times(M+1)}$ defined by:

1. $d_{0,0} = 0$;
2. $d_{i,0} = \infty$ for $i \in \{1, \cdots, N\}$;
3. $d_{0,j} = \infty$ for $j \in \{1, \cdots, M\}$;
4. $d_{i,j} = c_{i-1,j-1} + \min\{d_{i-1,j-1}, d_{i-1,j}, d_{i,j-1}\}$ for $i \in \{1, \cdots, N\}$ and $j \in \{1, \cdots, M\}$.

Once the global distance matrix is built, the optimal path π is found, the corresponding cumulative distance $D'_{DTW}(X,Y) =: sim_{DTW}(X,Y)$ defined as sum of elements of C for the path π, it can be utilized as score to describe the fitness between the testing trajectory and the training trajectory. Differently from traditional point wise distance measures as Euclidean, Manhattan and Chebyshev distances, the DTW quantifies the degree of fitness achievable by stretching or compressing the time series X and Y in regard to time, allowing degradation patterns to be matched in spite of time difference.

4.2. Similarity based on Euclidean distance with dimensionality reduction by an Unsupervised Autoencoder - UAE

Firstly, the proposed method trains a predictive model in the latent space of an Unsupervised Autoencoder. UAE has a symmetric structure of encoding and decoding having n_h layers each ones. Each layer is based on a convolutional model with a weight matrix, a bias vector, and an activation function. In this case, the activation function of each layer in this study is set the same as usual. The proposed UAE structure is an adaption of the model presented by [3], its shown in the Figure 2.

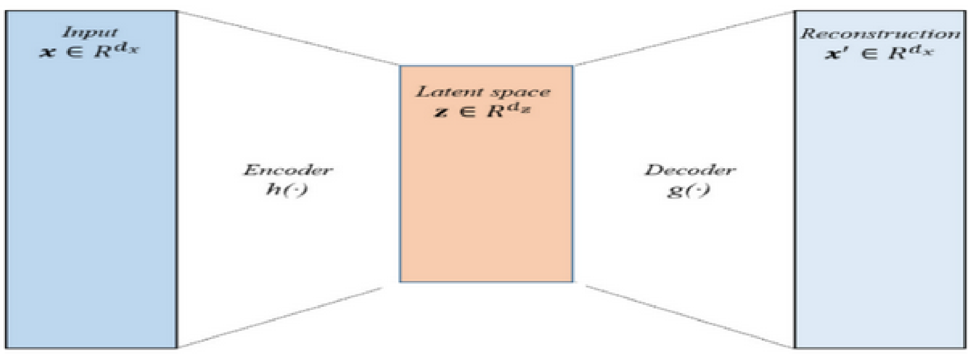

Figure 2: Unsupervised autoencoder under consideration.

In this study, $n_h = 2$ and the reconstruction loss is adopted as mean square error, also the activation function and optmizer are Leaky ReLU and Adam, respectively. After learning the UAE, it excluded the decoder, because it only required the latent vectors, calling the encoder as F.

Given a distance function d', a function F (i.e., the encoder of the model) and two time series, $X = (x_{t_0}, x_{t_1}, \cdots, x_{t_{N-1}})$, $N \in \mathbb{N}$ and $Y = (y_{t_0}, y_{t_1}, \cdots, y_{t_{M-1}})$, $M \in \mathbb{N}$, a similarity measure can be defined as:

$$sim_{UAE}(X, Y) := \frac{1}{1 + d'(F(X), F(Y))} .$$

Remark:

1. Note that the smaller $sim_{DTW}(X, Y)$, then X and Y would be more similar. In other hand, with the $sim_{UAE}(X, Y)$ occurs the opposite behavior.

4.3. Similarity Learning

As mentioned by [4], traditional deep learning models, such as CNN, might not work well when the training dataset is limited and unbalanced, and the working conditions deviate from those of the training dataset. Siamese and triplet network can achieve better performance under such conditions, being the main references in the similarity learning . In this section, adopt a distance d' and network F which would be optimized.

4.3.1. Siamese Network

Siamese network, often called twin network, consist of three main objects: a similarity function (e.g., the distance d'), a network (e.g., the network F) and a pair of instances as input to train it. The main objective is to identify whether two instances are similar or not, basically the distance between F applied two similar samples is minimized, and while the distance between F applied dissimilar ones is maximized. The way to obtain this behavior is minimizing the contrastive loss $\mathcal{L}_S$ associated the F which defined by

$$\mathcal{L}_S := (1 - Y) \cdot d'(F(x^{P1}), F(x^{P2})) + (Y) \cdot \max\{0, \alpha - d'(F(x^{P1}), F(x^{P2})\} \ ,$$

where x^{P1} and x^{P2} represent two different samples select randomly in a pair of inputs; Y is an indicator that takes the values of 0 if a pair of inputs have the same label, and 1 otherwise; α is a margin value that separate similar and dissimilar samples.

4.3.2. Triplet Network

The Triplet network is proposed as an improvement to the inherent problems of the Siamese network (i.e., samples closest the dissimilar and similar samples), having a similar objective and structure, varying only the separation of the dataset and the loss function. The structure of train consist of a triplet unit that contains an anchor sample, a positive sample whose label is the same as the anchor, and a negative sample with a different label to the anchor, seen the Figure 3 extracted of [3]. In this network, the distance between F applied the anchor and positive samples is minimized, simultaneously the distance between F applied the anchor and negative samples is maximized. The way to obtain this behavior is minimizing the triplet loss $\mathcal{L}_T$ associated the F which defined by

$$\mathcal{L}_T := \max\{0, d'(F(x^a), F(x^p)) - d'(F(x^a), F(x^n)) + \alpha\} \ ,$$

where x^a, x^p and x^n represent the anchor sample, positive sample and negative sample, respectively; α is a margin value that constructs a boundary between the classes.

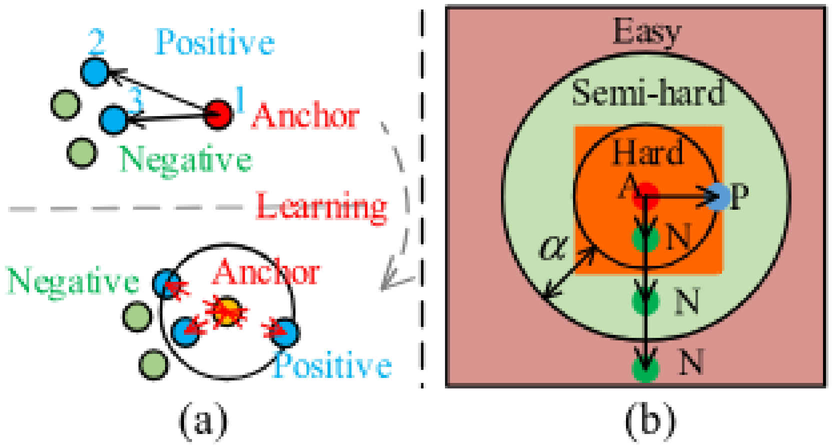

Figure 3: Training process of triplet network. (a) Coupled cluster loss. (b) Triplet units.

Remark:

1. In all cases, d' is adopted as the Euclidean distance;
2. In the similarity learning, α depends on the specialist's experience to select the adequate one. In this work, it adopted $\alpha = 0.0075$ in both networks.

5. Ways of classifier

In the classification, the dataset must be randomly split into the support set and the evaluated set and then normalized. The principal objective this step is predict the targets to evaluated set based on targets from the support set. For each element X of the evaluated set obtains similarity for each element of the support set. So, T most similar elements of the support set are selected, $Y_1, \cdots, Y_T$ with targets $H_1, \cdots, H_T$.

5.1. Classifier by adapted k-nearest neighbors (KNN)

In this classifier, the estimated target of X is assigned by to class most common among its k nearest neighbors such as a plurality vote system of its neighbors.

5.2. Classifier by weighted combination

In this classifier, the estimated target of X is approximately adopt as a weight combination of the T most similar as:

$$H = \sum_{m=1}^{T} w_m H_m \quad ,$$

where weighting coefficient w_m was calculated according with:

$$w_m = \frac{\frac{1}{sim(X,Y_m)}}{\sum_{j=1}^{T} \frac{1}{sim(X,Y_j)}} \quad .$$

Remark:

1. k and T are positive integers, typically small.
2. As $sim_{DTW}(X, Y_j)$ for any $j \in \{1, \cdots, T\}$ could be 0, it added a small value of 0.0001 to every similarity involving DTW to avoid division by 0;
3. In the expression of the weighting coefficient to similarity based on Euclidean distance, adopt $sim(X, Y_j)$ instead of $\frac{1}{sim(X,Y_j)}$ for any $j \in \{1, \cdots, T\}$.

6. Dataset and experiment setting

The dataset addresses the condition assessment of a hydraulic test rig based on multi sensor data with 2205 samples, including 17 sensors distributed throughout system with four fault types and one stable flag. Any missing values aren't present in any raw field.

All fields of frequency of 1 Hz was select to predict the cooling condition and the stable flag which means 8 sensors. In the dataset, the cooling condition assume 3 different percentage values: close to total failure (3%), reduced efficiency (20%) and full efficiency (100%) would be converted by values uniformly distributed 1, 2 and 3, respectively. Also, the classification of stable flag adopt binary values: conditions were stable (0) and static conditions might not have been reached yet (1).

7. Evaluation results

In this study, it selected the data split 8-fold cross-validation for all proposed models, except the DTW. This model is a classic one that doesn't involve machine learning - ml, as well as, it required a time of execution extremely large, this way, it indicates the data split hold out with a reduction in the samples space. Also, it selected the following metrics for all models: accuracy, precision, recall and f1-score. In the models involving machine learning, it variates the parameters: number of the neighbors or margin with the aim of optimizing the results by each normalization presented. In the table below, it shows the separation of the dataset by each model.

Separation of the dataset			
Model	Support Set - %	Test Set - %	Train and Validation Set - %
DTW	10.00	20.00	-
UAE	-	20.00	80.00
Siamese	13.60	20.00	66.40
Triplet	13.60	20.00	66.40

In the model involving UAE, the train and validation set is used as support set to evaluate each sample of the test set (i.e., samples to be evaluated), thus it's not selected initially a set for this purpose. Note that all ml models shown in the table, the same test set proportion was maintained, thus providing a reliable comparison between the models involving ml. In the classic one (i.e., DTW), it doesn't need a train and validation set. Also, by processing reasons, the proportion of the support set and test set doesn't total samples set in full. Furthermore, the number of neighbors is fixed at three.

As we have different ml models, the number of epochs large enough for convergence of each model isn't equal. In this case, the number of epochs for the models UAE, Siamese and Triplet was 50, 100 and 100, respectively. In this experiment, it used the classifier by weighted combination for the DTW and UAE models and the classifier adapted by KNN in the others, based on references [3, 4].

In the tables below, they show the optimized average and maximum accuracy and f1-score by the number of neighbors for the cooling condition and stable flag by each model and normalization shown.

Sequence of Classification for Cooling Condition (mean|max)				
Model	Norm.	N. neigh	Accuracy-%	F1-score-%
DTW	max-min	3	100.00|100.00	100.00|100.00
	[-1, 1]	3	97.78|97.78	97.78|97.78
	dec. scal.	3	100.00|100.00	100.00|100.00
	z-score	3	100.00|100.00	100.00|100.00
UAE	max-min	5	99.35|100.00	99.35|100.00
	[-1, 1]	3	98.50|99.55	98.50|99.55
	dec. scal.	4	99.06|100.00	99.06|100.00
	z-score	3	99.69|99.77	99.69|99.77
Siamese	max-min	3	99.46|100.00	99.46|100.00
	[-1, 1]	2	98.98|100.00	98.98|100.00
	dec. scal.	4	96.58|97.72	96.56|97.71
	z-score	1	99.84|100.00	99.84|100.00
Triplet	max-min	2	99.10|100.00	99.10|100.00
	[-1, 1]	3	98.90|100.00	98.90|100.00
	dec. scal.	3	99.02|99.67	99.02|99.67
	z-score	2	99.47|100.00	99.47|100.00

Sequence of Classification for Stable Flag (mean | max)				
Model	Norm.	N. neigh	Accuracy-%	F1-score-%
DTW	max-min	3	82.22 | 82.22	82.39 | 82.39
	[-1, 1]	3	73.33 | 73.33	73.33 | 73.33
	dec. scal.	3	93.33 | 93.33	93.27 | 93.27
	z-score	3	95.56 | 95.56	95.56 | 95.56
UAE	max-min	5	86.28 | 90.48	86.34 | 90.51
	[-1, 1]	3	89.40 | 95.01	89.37 | 94.98
	dec. scal.	4	95.27 | 97.51	95.24 | 97.49
	z-score	3	96.29 | 97.73	96.26 | 97.72
Siamese	max-min	3	85.20 | 88.66	85.45 | 88.76
	[-1, 1]	2	80.09 | 84.04	80.35 | 84.13
	dec. scal.	4	84.36 | 89.58	84.79 | 89.73
	z-score	1	91.40 | 92.86	90.97 | 92.60
Triplet	max-min	2	88.31 | 90.88	88.52 | 91.03
	[-1, 1]	3	78.99 | 86.64	79.29 | 86.76
	dec. scal.	3	88.60 | 90.88	88.69 | 90.85
	z-score	2	90.11 | 92.83	90.29 | 92.93

The following section presents two key visualizations to evaluate the performance of the predictive model for cooling conditions. Firstly, the following graph is displayed to compare the actual values and the predicted values of the UAE model. In addition, the matrix confusion is provided.

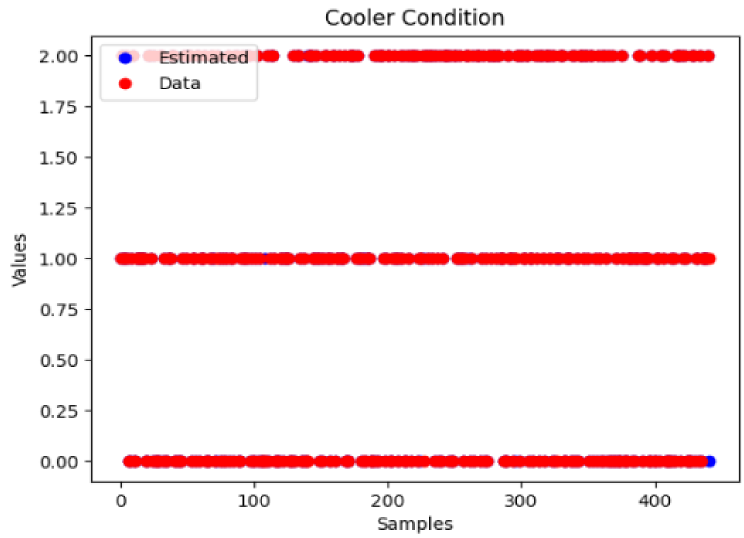

Figure 4: Actual value and UAE model prediction for Cooling Condition.

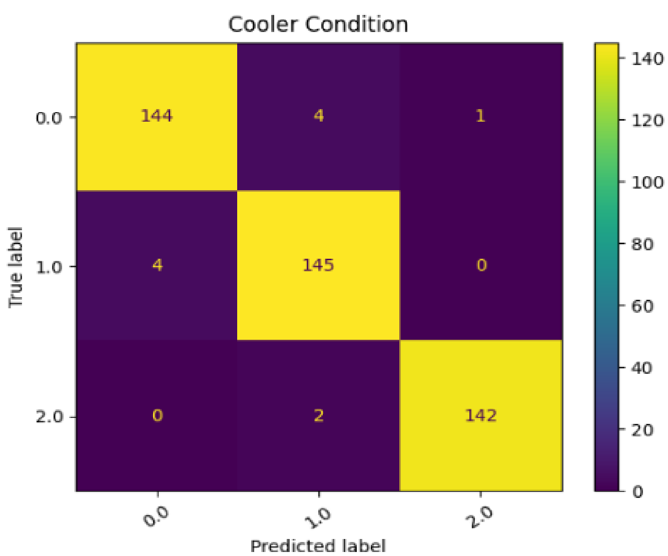

Figure 5: Confusion matrix for UAE model predictions for Cooling Condition.

Analogously, the comparative graph and the matrix confusion for the stable flag are shown for the UAE model.

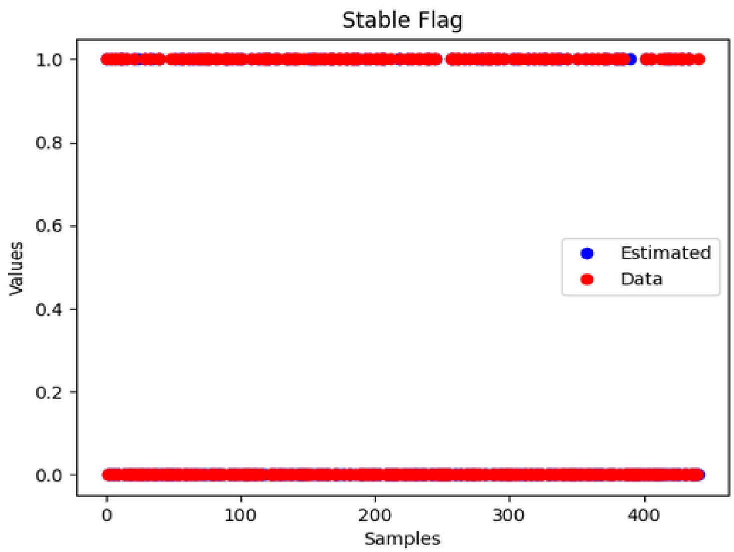

Figure 6: Actual value and UAE model prediction for Stable Flag.

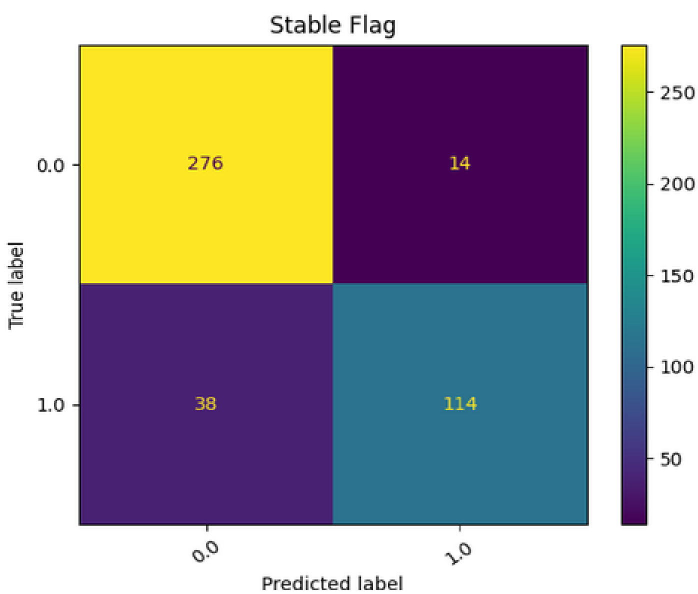

Figure 7: Confusion matrix for UAE model predictions for Stable Flag.

8. Discussion

Firstly, note that the number of neighbors influences the performance, and the ideal number of neighbors (i.e., the one that maximizes the performance) depends not only on the model, however, also on the type of data normalization.

By conducting a comparative analysis of the ml models shown, all of them demonstrate strong performance. In fact, all models achieved performance values (i.e., accuracy and F1-score) above 96% and 78% for cooling condition and stable flag, respectively. This highlights the potential of UAE, Siamese, and Triplet models as effective tools for similarity evaluations.

Additionally, note that UAE has proven better generalization capability. In fact, it has shown superior performance in 3 out 4 types of normalization, which are global [-1, 1], local decimal scaling and local z-score normalization. The UAE model with z-score normalization using 3 neighbors achieved the best performance. It achieved performance metrics exceeding 99% and 96% for cooling condition and stable flag, respectively.

By conducting a comparative anlysis of all models, note that the classical method (i.e., DTW) with global max-min normalization and global [-1, 1] normalization demonstrated inferior performance compared to the ml models shown. However, with the other normalizations, it demonstrated better performance than any similarity learning model used. Nevertheless, it still did not outperform the UAE. Furthermore, when comparing with the other models, the complexity of calculating the similarity between two samples is very costly using DTW. In fact, its complexity is $\mathcal{O}(n^2)$, while for the other models, after training, it is only $\mathcal{O}(n)$, which demonstrates the infeasibility of this model in scenarios where decisions need to be made quickly.

9. Conclusion

In summary, the comparative analysis conducted on the different similarity measures: DTW, UAE, Siamese and Triplet, along with the four distinct normalization techniques, demonstrate strong overall performance. The study indicates the influences of the number of neighbors and normalization in the performance of the model. Also, notably, the UAE model exhibited superior generalization capability in comparison with all four models shown, while still maintaining an acceptable level of algorithmic complexity.

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